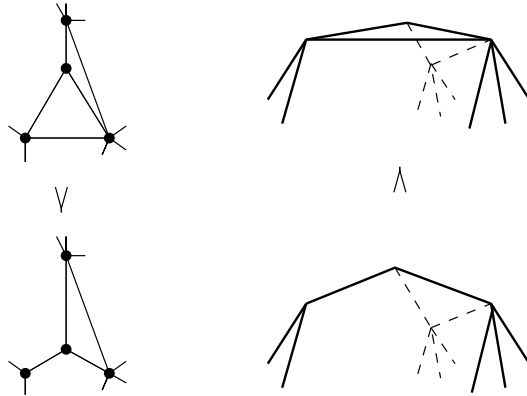


Proof. By duality, respectively polarity, we have to treat only the four types of Δ -to- Y transformations. For these, the transformation from P' to P just corresponds to “cutting off a vertex” by some suitable plane.

(To visualize this, consider our sketch of the four types of simple Δ -to- Y transformations, interpreting them as pictures of 3-polytopes.)



4.3 Planar Graphs are ΔY Reducible

In this section we show that every 3-connected planar graph (with $n \geq 4$ edges) can be reduced to K_4 by a sequence of simple ΔY transformations. This is a special case of a much more powerful theorem (for 2-connected planar graphs plus a “return edge”) that was first established by Epifanov [198] and has a clever and simple proof by Truemper [546]. In this section, we follow his expositions in [546] and [547, Sect. 4.3].

For a while, we will only require that the graphs considered are 2-connected; in particular, we admit ΔY operations if they keep our graphs 2-connected.

In the simplicity of Truemper’s approach to Epifanov’s theorem, the reader should appreciate the “power of a normal form theorem” (in this case: the embedding of a planar graph as a minor of a grid graph, which is also at the heart of Robertson & Seymour’s work [464] on graph minors).

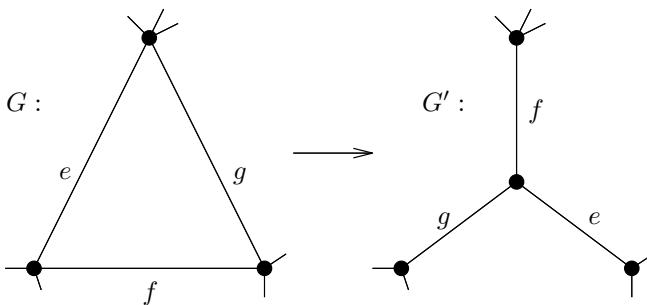
Here come three lemmas and a corollary, which together prove everything. [Working through this, you can practice “three levels of reading”: first read only the lemmas, and try to understand what they mean and how much is trivial. Second glance over the proofs, and try to see whether you know how to do them yourself. If you know, just do it. If you don’t, try to find counterexamples. In the third step, work your way through the proofs until you are confident that you found all the errors I made and the shortcuts I missed. Tell me about them.]

For the following, we will call a 2-connected graph G ΔY -reducible if it can be transformed into the graph C_2 with two parallel edges by a sequence of ΔY transformations and SP reductions.

Lemma 4.4. *If a planar graph G is ΔY -reducible, then so is every 2-connected minor H of G .*

Proof. We use an induction on the number of reduction steps that are necessary to reduce G . We can assume that H has no series or parallel edges, otherwise we can make the corresponding reductions. Now if the reduction for G starts with a series-parallel reduction step, then H is a minor of the reduced graph as well, because it does not contain both of the series or parallel edges.

Therefore, we can assume that the reduction of G starts with a ΔY reduction step. Exploiting duality, we may assume that this is a Δ -to- Y step $G \longrightarrow G'$.



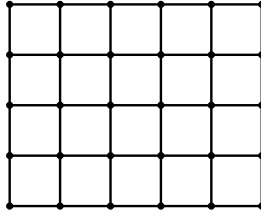
Let e, f, g be the three edges of G that are involved. One possibility is that all three edges are contained in H : then they form a nonseparating triangle in H as well, and we can perform the corresponding Δ -to- Y step $H \longrightarrow H'$. Then by induction we get that H' is ΔY -reducible, and hence so is H .

In the other case, some of the three edges e, f, g do not appear in H . What happened to them? Since H is simple, it is not possible that only one or two of them were contracted. If all three were contracted, then we can assume that first one edge was deleted, then the others were contracted. Using that deletions and contractions commute, and possibly relabeling, we can thus assume that the first edge that disappears when H is formed from G , say e , is deleted.

But then we get the same minor H from G' , by contracting the corresponding edge from G' , because the deletion of e from G and the contraction of e in G' result in the same graph. Again we are finished by induction. $\square$

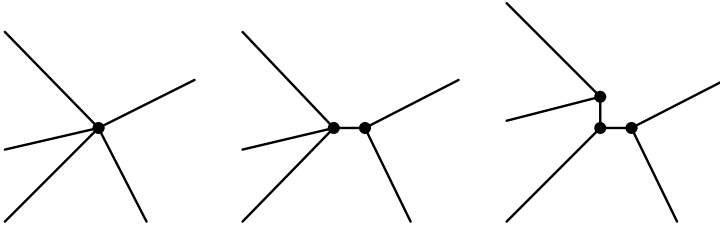
Denote the *grid graph* with mn vertices and $m(n-1) + n(m-1)$ edges by $G(m, n)$. Clearly the grid graphs $G(m, n)$ and $G(n, m)$ are isomorphic.

Our sketch shows the grid graph $G(5, 6)$.

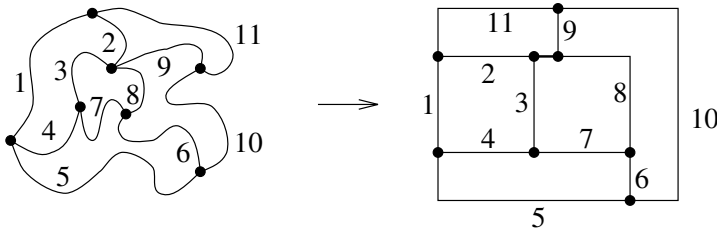


Lemma 4.5. *If G is planar, then it is a minor of a grid graph.*

Proof. For this, fix an embedding of G into the plane $\mathbb{R}^2$; now split the vertices of G in order to get a graph G' of which G is a minor, such that all vertices of G have degree at most 3.



Then we can construct an embedding of G' (with subdivided edges) into a finite grid that is combinatorially equivalent to the given embedding (in the sense that the vertices at a given edge come in the same cyclic order). For that we draw one edge at a time on the grid. Whenever our grid is too coarse, we can refine it by taking half the grid size.

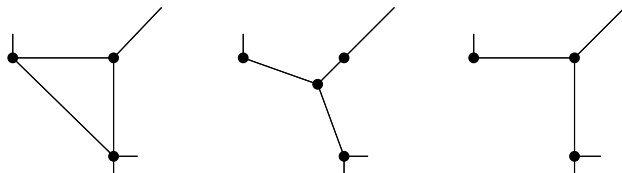


Once a subdivision graph G' is represented as a subgraph of the grid graph this way, it is clear that G is a minor of the grid graph. $\square$

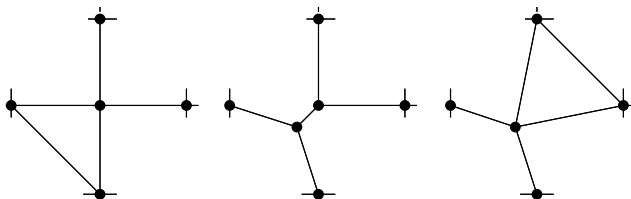
This is quite trivial. Much more sophisticated versions of this, restricting the size of the grid we embed into, are interesting and important for applications in VLSI-layout problems. We refer to Lawler et al. [350, Sect. 3.4.5], Lengauer [359, Ch. 5], and Kant [311] for further information as well as more precise and sophisticated versions of such “grid layout” and “graph drawing” results.

Lemma 4.6. *All grid graphs $G(m, n)$ with $m, n \geq 3$ are ΔY -reducible to K_4 .*

Proof. We use two basic observations. First, if an edge connects two neighbors of a vertex of degree 3, then we can delete it, by performing first a Δ -to- Y transformation and then a series reduction.

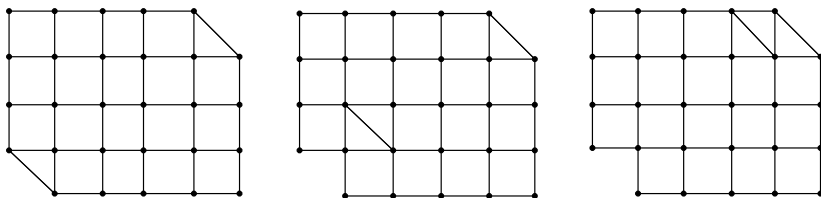


Second, if an edge connects two neighbors of a vertex of degree 4, then we can move the edge over to “the other side,” using first a Δ -to- Y transformation and then a Y -to- Δ transformation.

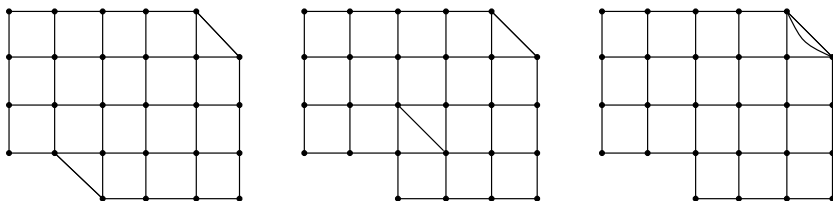


Using these two observations, we can reduce any grid graph $G(m, n)$ to K_4 , as follows.

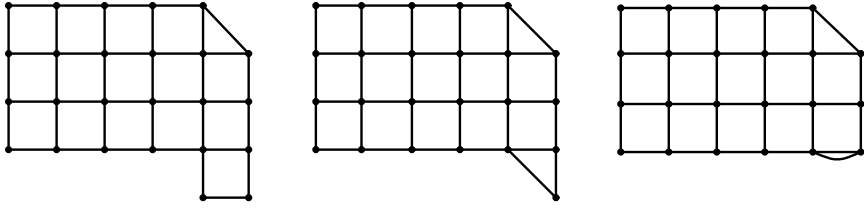
First perform a series reduction in the upper-right corner. Then, assuming that $m \geq 4$, we take the two edges in the lower-left corner and perform a series reduction to get a single edge. We can move this single edge across the whole grid in a sequence of degree-4 moves until we “hit the boundary.”



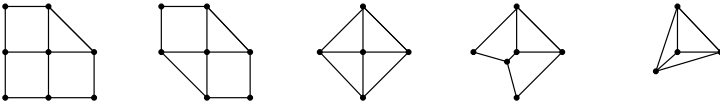
There either the edge is parallel to the diagonal corner edge and we can parallel-delete, or it can be deleted according to our degree-3 move.



This way we have deleted the first square in the last row; similarly, we can delete the second square in this row, and so on; the last square is deleted by two series reductions and one parallel reduction.



We can delete all the squares in the first column of our board symmetrically, if $n \geq 4$.



At the end, this leaves us with the grid $G(3, 3)$ with one short corner, which is easily reduced first to the “wheel” graph W_4 of a square pyramid, and then to K_4 . $\square$

Corollary 4.7. *Every 3-connected planar graph G can be reduced to K_4 by a sequence of simple ΔY transformations.*

Proof. The three lemmas together show that G is ΔY -reducible. We follow this reduction to the first point where parallel or series edges are created. These can be reduced immediately, which also results in a 3-connected planar graph G by Lemma 4.2. The graph we now have has fewer edges; hence we are done by induction on size. $\square$

Corollary 4.7 also completes our proof of Steinitz' Theorem 4.1.

4.4 Extensions of Steinitz' Theorem

Corollary 4.8. *For every 3-polytope, there is a rational 3-polytope that is combinatorially equivalent.*

Proof. This follows from our proof of Steinitz' theorem: we have shown in fact that every 3-polytope can be reduced to a simplex by two types of operations $P \longrightarrow P'$:

- constructing a combinatorially polar polytope,
- “cutting off” a vertex of degree 3.

In both cases, if we have a rational realization of P' , then we can also construct P with rational vertices and inequalities. $\square$